



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 02/02/2023*

PROBLEM - We observe two samples of the discrete-time real-valued w.s.s. random process $S[n]$ at $n=0$ and $n=2$, where $S[n]$ is a w.s.s. zero-mean Gaussian process modelled as an Auto-Regressive (AR) process of order 1, i.e. $S[n] = \rho S[n-1] + I[n]$, where ρ is the one-lag correlation coefficient of the process $S[n]$ and $I[n]$ is the white innovation process with power σ_I^2 .

(1) Derive the expressions of the power σ_S^2 , the autocorrelation function (ACF) $R_S[m]$ and the Power Spectral Density (PSD) $S_S(e^{j2\pi f})$ of the process $S[n]$.

(2) Derive the covariance matrix of the 2-D observed vector $\mathbf{S} = [S[0] \ S[2]]^T$ and its probability density function (PDF).

(3) Solve the following **missing data** problem: Assuming that for some reason we could not observe $S[1]$, find the optimal (in the mean square sense) linear interpolator that retrieves (estimates) $S[1]$ based on the observation of the two adjacent samples $S[0]$ and $S[2]$.

(4) Derive the bias $bias = E\{S[1] - \hat{S}[1]\}$ and the *Mean Square Error* (MSE) $\sigma_\epsilon^2 \triangleq E\{(S[1] - \hat{S}[1])^2\}$ of the optimal linear estimator $\hat{S}[1]$; plot the MSE versus ρ , analyse what happens to $\hat{S}[1]$ and σ_ϵ^2 when $\rho \rightarrow -1$, $\rho \rightarrow 0$, and $\rho \rightarrow 1$ and comment the results.



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Solution

(1) Power, ACF and PSD.

$$S[n] = \rho S[n-1] + I[n]$$

$$S(z) = \rho z^{-1} S(z) + I(z) \Rightarrow L(z) = \frac{S(z)}{I(z)} = \frac{1}{1 - \rho z^{-1}} = \frac{z}{z - \rho}$$

The innovation filter has 1 zero at $z = 0$ and 1 pole at $z = \rho$. Hence, the innovation filter is a high-pass filter if $\rho < 0$ and a low-pass filter if $\rho > 0$, and so the process. The ACF and the PSD are given by (see the course vugraphs for the derivation):

$$R_s[m] = E\{S[n]S[n+m]\} = \frac{\sigma_I^2}{1 - \rho^2} \rho^{|m|}, \text{ hence } \sigma_s^2 = R_s[0] = \frac{\sigma_I^2}{1 - \rho^2}.$$

$$S_s(e^{j2\pi f}) = \sigma_I^2 \left| L(e^{j2\pi f}) \right|^2 = \sigma_I^2 \left| \frac{1}{1 - \rho e^{-j2\pi f}} \right|^2 = \frac{\sigma_I^2}{1 + \rho^2 - 2\rho \cos(2\pi f)}$$

As previously stated, when $\rho < 0$, $S[n]$ is a high-pass process, when $\rho > 0$, $S[n]$ is a low-pass process.

(2) Covariance matrix of the data vector $\mathbf{S}$.

The covariance matrix $\mathbf{C}_s = E\{\mathbf{S}\mathbf{S}^T\}$ is as follows:

$$\mathbf{C}_s = \begin{bmatrix} R_s[0] & R_s[-2] \\ R_s[2] & R_s[0] \end{bmatrix} = \sigma_s^2 \begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix} = \sigma_s^2 \mathbf{M}$$

where $\mathbf{M}$ is the normalized covariance matrix. The pdf of $\mathbf{S}$ can be derived by observing that $S[n]$ is a Gaussian process, so $\mathbf{S}$ is a Gaussian vector:

$$\mathbf{S} \in N(\boldsymbol{\eta}_s, \mathbf{C}_s), \text{ where } \boldsymbol{\eta}_s = E\{\mathbf{S}\} = \mathbf{0} \text{ and } \mathbf{C}_s = \sigma_s^2 \begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix} = \sigma_s^2 \mathbf{M}.$$

(3) LMMSE estimator

Since the process is Gaussian, the LMMSE coincides with the MMSE and the MAP estimators.

The LMMSE estimator is obtained by applying the Orthogonality Principle:

$$\hat{S}[1] = a_1 S[0] + a_2 S[2] + b, \quad \varepsilon[1] = S[1] - \hat{S}[1]$$

Since $S[n]$ is a zero mean process we have that: $E\{\varepsilon[1]\} = E\{S[1] - \hat{S}[1]\} = 0 \Rightarrow b = 0.$



$$E\{\varepsilon[1]S[1+m]\} = 0, \quad m = -1, 1$$

$$E\{(S[1] - \hat{S}[1])S[1+m]\} = 0, \quad m = -1, 1$$

$$E\{(S[1] - a_1 S[0] - a_2 S[2])S[1+m]\} = 0, \quad m = -1, 1$$

$$R_s[m] - a_1 R_s[m+1] - a_2 R_s[m-1] = 0, \quad m = -1, 1$$

$$\begin{cases} a_1 R_s[0] + a_2 R_s[-2] = R_s[-1] \\ a_1 R_s[2] + a_2 R_s[0] = R_s[1] \end{cases}$$

$$\begin{bmatrix} R_s[0] & R_s[-2] \\ R_s[2] & R_s[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} R_s[-1] \\ R_s[1] \end{bmatrix}$$

$$R_s[0] = \sigma_s^2, \quad R_s[1] = R_s[-1] = \sigma_s^2 \rho, \quad R_s[-2] = R_s[2] = \sigma_s^2 \rho^2$$

$$\sigma_s^2 \begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \sigma_s^2 \begin{bmatrix} \rho \\ \rho \end{bmatrix}$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \rho \begin{bmatrix} 1 & \rho^2 \\ \rho^2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\rho}{1-\rho^4} \begin{bmatrix} 1 & -\rho^2 \\ -\rho^2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\rho(1-\rho^2)}{1-\rho^4} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{\rho}{1+\rho^2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{S}[1] = \frac{\rho}{1+\rho^2} S[0] + \frac{\rho}{1+\rho^2} S[2] = \frac{2\rho}{1+\rho^2} \cdot \left(\frac{S[0] + S[2]}{2} \right)$$

(3) Bias and MSE estimator

$$E\{\varepsilon[1]\} = E\{S[1] - \hat{S}[1]\} = 0 \Rightarrow \text{bias} = 0$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[1]\} = E\{(S[1] - \hat{S}[1])S[1]\} = E\left\{\left(S[1] - \frac{\rho}{1+\rho^2} S[0] - \frac{\rho}{1+\rho^2} S[2]\right)S[1]\right\} \\ &= R_s[0] - \frac{\rho}{1+\rho^2} R_s[1] - \frac{\rho}{1+\rho^2} R_s[-1] = \sigma_s^2 - \frac{2\rho}{1+\rho^2} \cdot \sigma_s^2 \rho \\ &= \sigma_s^2 \left(1 - \frac{2\rho^2}{1+\rho^2}\right) = \frac{1-\rho^2}{1+\rho^2} \sigma_s^2 \end{aligned}$$



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In summary:

$$\hat{S}[1] = \frac{2\rho}{1+\rho^2} \cdot \left(\frac{S[0]+S[2]}{2} \right), \quad \sigma_\varepsilon^2 = \frac{1-\rho^2}{1+\rho^2} \sigma_s^2$$

$$\rho \rightarrow -1: \hat{S}[1] \rightarrow -\frac{S[0]+S[2]}{2}, \quad \sigma_\varepsilon^2 = 0$$

parameter to be estimated and observed data fully correlated.

$$\rho \rightarrow 0: \hat{S}[1] \rightarrow 0 = E\{S[1]\}, \quad \sigma_\varepsilon^2 = \sigma_s^2 = \text{var}\{S[1]\}$$

parameter to be estimated and observed data uncorrelated.

$$\rho \rightarrow 1: \hat{S}[1] \rightarrow \frac{S[0]+S[2]}{2}, \quad \sigma_\varepsilon^2 = 0$$

parameter to be estimated and observed data fully correlated.

